Vicente Hojas

PhD Student in Applied and Computational Mathematics
California Institute of Technology
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Education

2024–Present **California Institute of Technology**
- PhD in Applied and Computational Mathematics. Advisor: Oscar P. Bruno. Expected graduation: 09/2030.

2017–2024 **Pontificia Universidad Católica de Chile**
- MSc in Mathematical Engineering (2021–2024). Advisors: Manuel Sánchez and Carlos Pérez-Arancibia (University of Twente).
- Professional Degree in Mathematical and Computational Engineering (2017–2024).
- BS in Engineering, Major in Mathematical Engineering (2017–2021). Ranked first among 479 students.

Research Experience

2024–Present **Doctoral Research**, *California Institute of Technology*

Advisor: Oscar P. Bruno.
- Develop high-order C++ boundary integral solvers for wave scattering by periodic structures, including geometries with sharp corners, complex frequencies, and Rayleigh–Wood anomalies.
- Implemented the interpolated factored Green function (IFGF) algorithm to accelerate wave-scattering computations from $\mathcal{O}(N^2)$ to $\mathcal{O}(N \log N)$ complexity.
- Compute scattering resonances of 2D and 3D objects using rational approximation of frequency-domain scattering responses.
- Develop resonance-based target identification methods using generalized likelihood ratio tests to classify objects from noisy measurements.

2021–2024 **Master's Research**, *Pontificia Universidad Católica de Chile*
- Developed reflectionless discrete perfectly matched layers (PMLs) for high-order finite difference schemes for wave propagation, eliminating numerical reflections caused by discretizing standard PMLs.
- Published the resulting method as first author in the *SIAM Journal on Scientific Computing*.

Publications

2024 **Vicente A. Hojas**, Carlos Pérez-Arancibia, and Manuel A. Sánchez. Reflectionless Discrete Perfectly Matched Layers for Higher-Order Finite Difference Schemes. *SIAM Journal on Scientific Computing*, 46(5), A3094–A3123. doi:10.1137/23M1581558.

Technical Skills

Programming C++, C, Python, MATLAB.

Parallel computing OpenMP, CUDA.

Methods Boundary integral equations, fast algorithms, high-order methods, finite differences, numerical linear algebra, and rational approximation.

Talks and Posters

Apr. 2026 **Target Identification of Complex 2D and 3D Geometries via Scattering Poles**, *Interactions of Geometric Spectral Theory with Numerical Methods and Applications, Isaac Newton Institute for Mathematical Sciences*, Cambridge, United Kingdom
Poster presentation. Joint work with Manuel A. Santana and Oscar P. Bruno.

Jan. 2024 **Reflectionless Discrete PMLs for High-Order Finite Difference Schemes and Finite Element Methods**, *WONAPDE 2024*, Concepción, Chile
Talk, Minisymposium on Computational Electromagnetism.

Mar. 2023 **Reflectionless Discrete Perfectly Matched Layers for Higher-Order Finite Difference Schemes**, *University of Twente*, Enschede, The Netherlands
Seminar, Mathematics of Computational Science group.

Mar. 2023 **Reflectionless Discrete-Analytic Perfectly Matched Layers for Higher-Order Finite Difference Schemes**, *SIAM Conference on Computational Science and Engineering (CSE23)*, Amsterdam, The Netherlands
Poster presentation.

Honors and Awards

2023 ANID National Master's Scholarship.

2023 SIAM Travel Award, CSE23.

2022 Academic Excellence Award for Interdisciplinary Major, Pontificia Universidad Católica de Chile.

2017 Adolfo Rojas Scholarship.

Teaching Experience

2025–Present **Teaching Assistant**, *California Institute of Technology*
ACM101a and ACM101b: Methods of Applied Mathematics; ACM104: Applied Linear Algebra. Held office hours and graded coursework.

2019–2023 **Teaching Assistant**, *Pontificia Universidad Católica de Chile*
- IMT2111: Numerical Linear Algebra. Led recitation sessions, held office hours, and graded coursework.
- IIC1253: Discrete Mathematics; MAT1630: Calculus III. Graded coursework.

Languages

Spanish (native), English.